

$$C = 2\% \text{ of } A$$

$$C = \frac{x}{100} \times \frac{y}{5}$$

$$x = 500$$

10)

DAY - 17

Date: 24/01/26

What percentage?

Formula 1:

$$x \text{ \% is what of } y = \frac{x}{y} \times 100$$

$$\text{what \% of } x \text{ is } y = \frac{y}{x} \times 100$$

Formula 2:

x is what % more (or) less than

$$y = \frac{x-y}{y} \times 100$$

+	-
more	less

1) If x is 10% more than y , then by what percent is y less than x ?

$$\begin{array}{cc} x & y \\ 110 & 100 \end{array}$$

$$= \frac{y-x}{x} \times 100$$

$$= \frac{100-110}{110} \times 100 = -\frac{10}{11} \times 100$$

$$\Rightarrow -\frac{100}{11} \Rightarrow -\frac{9}{11}$$

$$\approx \frac{9}{11}$$

2) IF A's height is 10% more than B's height, by how much percent less is B's height than that of A?

$$\begin{array}{cc} A & B \\ 110 & 100 \end{array}$$

Than/of

$$= \frac{B-A}{A} \times 100 = \frac{100-110}{110} \times 100$$

$$= -\frac{10}{11}$$

3) B got 20% marks less than A.

what marks did A got more than B?

$$\begin{array}{cc} A & B \\ 100 & 80 \end{array}$$

$$= \frac{A-B}{B} \Rightarrow \frac{100-80}{80} \times 100$$

$$= \frac{\frac{20}{20}}{4} \times 100 \Rightarrow \frac{100}{4} \Rightarrow 25\%$$

4) If x earns 25% more than y.

what percent less does y earn

than x?

$$\begin{array}{cc} x & y \\ 125 & 100 \end{array}$$

$$= \frac{y-x}{x} \times 100 \Rightarrow \frac{100-125}{125} \times 100$$

$$= -\frac{25}{125} \times 100 \Rightarrow -20\%$$

$$= 20\%$$

5) Two numbers are respectively $12\frac{1}{2}\%$ and 25% more than a third number. The first number is what percentage of second number is:

x	y	z
$12\frac{1}{2}\%$	25%	?
$\Downarrow$	$\Downarrow$	$\Downarrow$
$\frac{225}{200}$	125	100

$$\Rightarrow \frac{25}{200} \times 100$$

$$2 \times \frac{25}{200} \times 100$$

$$\Rightarrow 100 + \frac{25}{2}$$

$$= \frac{225 - 125}{125 \times 2} \times 100$$

$$= \frac{225}{5} \Rightarrow 45 \times 2$$

$$= 90\%$$

$$12\frac{1}{2}\%$$

$$\frac{25}{2}\%$$

6) Two numbers are less than a third number by 30% and 37% respectively. The percent by which the second number is less than the first is

X	Y	Z
↓	↓	↓
- 30%	- 37%	100
↓	↓	↓
70	63	100

$$= \frac{63 - 70}{70} \times 100$$

$$= \frac{7}{70} \times 100$$

$$= \boxed{10\%}$$

$$\begin{array}{r} 100 \\ - 37 \\ \hline 63 \end{array}$$